Toward a four-dimensional Yang–Mills construction

Notebook 05: the flat gauge bridge and cutoff interactions

Research working document

16 October 2026

Abstract

This note proves two bridges needed to apply the earlier Gaussian estimates to the lattice construction. First, it identifies the actual averaged-path gauge penalty, distinguishes the rooted and block-mean constraints, and transfers a flat axial gap quantitatively to the exponential gauge. Second, physical-field analyticity gives the correct enlarged-cutoff bounds for the interaction, including the linear Jacobian term. A complete Gaussian cluster argument is given for interactions within disjoint fixed-dimensional cells, with an explicit bound on the change of the logarithm. The actual Yang–Mills interaction joins overlapping cells; its integration theorem is still a separate, unverified interface. No continuum theory or mass gap is asserted.

Status and sources. These are internally checked finite-dimensional and conditional estimates, without external mathematical review or a novelty claim. Raw I and Raw II refer to the pinned deposited account. The model formulas are compared with the original articles [1, 2, 3]. Notebook 03 supplies the Gaussian forest and logarithm estimates; Notebook 04 supplies the stronger local-precision support algebra. Their hypotheses are retained whenever invoked below.

1 The actual flat gauge penalty and its constraint

Work on a unit lattice partitioned into blocks of side L, with $n = L^d$ sites and a distinguished center y in each block $B(y)$. Count one positive orientation per bond. Lie-algebra coordinates are Hermitian, with invariant norm $|X|^2 = \operatorname{tr}(X^2)$ in a fixed normalization. The following statements use the ordinary block tiling with distinct interfaces. A compatible Dirichlet convention takes a region made of complete blocks, retains bonds with at least one endpoint in it, sets fields to zero on bonds with both endpoints outside, and sets gauge parameters to zero outside. Blockwise mean subtraction then preserves the allowed site space and the boundary pivots remain available. This is the incident-bond convention in Raw II, line 9033 and the positivity hypothesis beginning at line 9743. Freezing all crossing bonds would require a different argument. Periodic seams must remain distinct geometric edges, even with only two blocks in a direction.

For each $x \in B(y)$, let $\mathcal{P}(y, x)$ be the prescribed family of monotone shortest paths. Define the averaged path integral

$$(TB)(y, x) = \operatorname*{average}_{\Gamma \in \mathcal{P}(y,x)} \sum_{b \in \Gamma} \epsilon_b B_b, \qquad (TB)(y, y) = 0.$$

Let T_0 denote a single rooted-tree path integral, choosing a tree whose root-to-site paths have length at most $h = d(L - 1)$. Write $D\lambda(v, w) = \lambda(w) - \lambda(v)$ and $d\lambda(y, x) = \lambda(x) - \lambda(y)$. Telescoping gives

$$TD\lambda = T_0 D\lambda = d\lambda. \tag{1}$$

On rooted site fields $\lambda(y) = 0$, this is $TD\lambda = \lambda$. The useful elementary bounds are

$$\|D\|_{2\to 2} \le d_0 := 2\sqrt{d}, \qquad \|T\|_{2\to 2}, \|T_0\|_{2\to 2} \le t_0 := \sqrt{h(n-1)}. \tag{2}$$

For D, sum $|\lambda(v) - \lambda(w)|^2 \le 2|\lambda(v)|^2 + 2|\lambda(w)|^2$. For T, each absolute coefficient row sums to at most h, and each column to at most $n-1$; the Schur test proves the assertion.

The gauge fixing in CMP109, equations (0.11), (0.14)–(0.16), uses these averaged contours. If its group variable is $\Omega(t) = \exp(i[tTB + O(t^2)])$, then

$$1 - \Re \operatorname{tr} \Omega(t) = \tfrac{1}{2} t^2 |TB|^2 + O(t^3).$$

Thus its actual quadratic penalty, with the stated normalization, is $\frac{1}{2}\|TB\|^2$. The second-order term in the logarithm does not contribute to this coefficient. This matches CMP109 (2.5). For a tree-axial field B_0, generally $TB_0 \neq 0$. Hence on $B_0 + D\lambda$ the penalty is $\frac{1}{2}\|TB_0 + \lambda\|^2$, requiring a shear of the gauge coordinate before it becomes diagonal.

1.1 Rooted and block-mean constraints are different

For a coarse bond $c = (y, y + Le_\mu)$, set

$$(Q_{\mathrm{st}}B)(c) = L^{-d-1} \sum_{x \in B(y)} \int_{[x, x+Le_\mu]} B,$$

$$(aTB)(y) = L^{-d} \sum_{x \in B(y)} (TB)(y, x), \qquad (D_1 f)(c) = L^{-1}[f(y + Le_\mu) - f(y)].$$

Linearizing the rooted contour average of CMP109 (0.12) gives

$$Q^\flat = Q_{\mathrm{st}} - D_1 aT, \qquad Q_1 := Q^\flat + D_1 aT = Q_{\mathrm{st}}. \tag{3}$$

Indeed the contour integral is the initial rooted path, plus the translated straight string, minus the final rooted path and the reference coarse string. Multiplication by the reference transporter cancels the last term. Averaging and dividing by L gives (3). In particular,

$$Q^\flat D\lambda = D_1(\lambda \text{ at centers}), \qquad Q_1 D\lambda = D_1(\text{block mean of } \lambda). \tag{4}$$

Rooted gauge directions preserve $\ker Q^\flat$; block-mean-zero directions preserve $\ker Q_1$. The two constraints agree on $\ker T$.

For $B \in \ker Q^\flat$, take the rooted field $\xi = TB$ and put

$$B_T = B - D\xi. \tag{5}$$

Then $TB_T = 0$, $Q^\flat B_T = Q_1 B_T = 0$. Conversely any $B_T \in \ker Q_1 \cap \ker T$ and any rooted ξ give a unique $B = B_T + D\xi \in \ker Q^\flat$ with $TB = \xi$.

To compare with a tree slice inside the correct constraint, use the block-mean-zero site space N. The inverse of $d : N \to \{f(y, y) = 0\}$ is

$$(d^{-1}f)(x) = f(y, x) - L^{-d} \sum_{z \in B(y)} f(y, z), \qquad \left\| d^{-1} f \right\|_2 \le \|f\|_2 .$$

The norm inequality is orthogonal subtraction of the block mean. The mutually inverse maps between the two axial slices are

$$B_0 = B_T - Dd^{-1}T_0 B_T, \qquad B_T = B_0 - Dd^{-1}TB_0. \tag{6}$$

They preserve Q_1 by (4); telescoping verifies the slice conditions and both inverse identities. In particular

$$\|B_T\| \le (1 + d_0 t_0) \|B_0\| . \tag{7}$$

2 A quantitative flat coercivity transfer

Theorem 2.1. *Let q be a real symmetric quadratic form on bond fields with $q(B + D\lambda) = q(B)$ for the relevant gauge directions. Suppose*

$$q(B_0) \geq \gamma_{\mathrm{ax}} \|B_0\|^2 \quad \text{on } \ker Q_1 \cap \ker T_0.$$

Then, on $\ker Q^\flat$,

$$q(B) + \|TB\|^2 \geq \gamma_{\mathrm{exp}} \|B\|^2, \qquad \gamma_{\mathrm{exp}} = \left(\frac{(1 + d_0 t_0)^2}{\gamma_{\mathrm{ax}}} + d_0^2 \right)^{-1}. \tag{8}$$

If the penalty is $a\|TB\|^2$ with $a > 0$, replace d_0^2 by d_0^2/a in the denominator.

Proof. Gauge invariance, (6), and (7) give $q(B_T) \geq \gamma_T \|B_T\|^2$, where $\gamma_T = \gamma_{\mathrm{ax}}/(1 + d_0 t_0)^2$. The exact decomposition (5) yields

$$q(B) + \|TB\|^2 = q(B_T) + \|\xi\|^2 \geq \gamma_T \|B_T\|^2 + \|\xi\|^2 .$$

Weighted Cauchy–Schwarz applied to $\|B\| \leq \|B_T\| + d_0 \|\xi\|$ gives (8). The version with a is identical. □

The axial input here concerns the full constrained bond norm. A statement in independent coordinates $B_0 = K_{\mathrm{ax}} z$ with $q(K_{\mathrm{ax}} z) \geq \gamma_{\mathrm{free}} \|z\|^2$ only gives $\gamma_{\mathrm{ax}} = \gamma_{\mathrm{free}} / \|K_{\mathrm{ax}}\|^2$. This conversion is essential.

2.1 Explicit flat pivot-coordinate bounds

Choose one pivot $b_0(c)$ for each coarse bond: the unit bond crossing its block interface on the line through the two centers. No rooted contour in T uses this interblock bond. Exactly L starting sites in the straight-string sum use it, so its coefficient in $Q^\flat$ is L^{-d}. No other coarse equation uses the pivot. Therefore, after splitting the remaining coordinates z,

$$B_{\mathrm{pivot}} = -L^d Q_f z, \qquad \|K_\flat\|^2 \leq 1 + L^{2d} \left\| Q^\flat \right\|^2 .$$

Put $r = 1 + 2h/L$. The absolute row sum of $Q^\flat$ is at most r. An internal bond can enter contour terms only through its own block, which has $2d$ incident coarse bonds; straight strings add at most two equations. Thus the absolute column sum is at most $(2d + 2)r$, and

$$\|K_\flat\|^2 \leq 1 + L^{2d}(2d + 2)r^2, \qquad \|K_{\mathrm{ax}}\|^2 \leq 1 + 2L^{2d}. \tag{9}$$

For the second bound, tree bonds vanish and $Q_1 = Q_{\mathrm{st}}$ has row sum one and column sum at most two; the same pivots remain free of the block trees. These conservative constants are independent of scale and volume for this tiling. A different pivot scheme or a boundary that identifies interfaces needs its own coefficient matrix.

If also $q(B_0) \leq M_q \|B_0\|^2$ on the axial slice, the projections above imply

$$q(B) + \|TB\|^2 \leq [M_q(1 + d_0 t_0)^4 + t_0^2] \|B\|^2 .$$

After $B = K_\flat z$, the lower bound remains $\gamma_{\mathrm{exp}} \|z\|^2$ since z is a coordinate restriction of B, and the upper bound gains $\|K_\flat\|^2$. Both bounds matter for a uniformly accretive complex covariance.

2.2 Why this applies at the flat reference, and its limit

For the actual gauge-invariant effective action $f(V) = \mathcal{A}(U_k(V))$, the trivial datum is stationary: the Wilson action is nonnegative and vanishes at the trivial field. The Hessian therefore annihilates infinitesimal gauge directions. The gradient-times-second-chart-derivative correction also vanishes there. Consequently the flat action Hessian has exactly the invariance needed in Theorem 2.1, and the actual gauge penalty is the one derived above. This identifies a bridge from a uniform axial estimate, not an independent proof of that axial estimate for every effective scale.

The algebraic distinctions already occur in Raw II, lines 9045–9285; the deposited account's later quantitative covariance statement has additional hypotheses. The remaining model inputs include the primary axial gap and upper bound in the matched coordinates, and uniform weighted perturbation estimates for the actual complex independent-current family. A change of coordinates in a coercivity proof does not authorize a change of the real sharp-cutoff integration domain.

3 The current correction to gauge invariance

Use $U'(b) = \exp(i\eta B_b)U_b$ and the convention $U^g(v,w) = g(v)^{-1}U(v,w)g(w)$. Expand the actual action as

$$f_U(B) = f(U) + \langle J_U, B\rangle + \tfrac{1}{2}H_U(B,B) + O(\|B\|^3).$$

For $g_t = \exp(it\lambda)$, write $a = \lambda(v)$, $b' = \mathrm{Ad}_{U(v,w)}\,\lambda(w)$ and $D_U\lambda = (b'-a)/\eta$. The Baker–Campbell–Hausdorff expansion of the transformed two-parameter field gives the mixed coordinate

$$sB + tD_U\lambda + \tfrac{i}{2}st[B, a+b'] + O(s^2t + st^2) + \text{terms independent of } s.$$

Gauge invariance therefore proves the exact Ward identity

$$H_U(B, D_U\lambda) = -\tfrac{i}{2}\langle J_U, [B, a+b']\rangle. \tag{10}$$

It also gives $\langle J_U, D_U\lambda\rangle = 0$. The pure-gauge second coordinate derivative is $-(i/\eta)[a,b']$, agreeing with $H_U(D_U\lambda, D_U\lambda) = (i/\eta)\langle J_U, [a,b']\rangle$ and checking the sign and scale.

For unitary U, choose c_{br} with $|[X,Y]| \le c_{\mathrm{br}}|X||Y|$, and put $j_* = \sup_b |J_U(b)|$. Since the square sum of $|\lambda(v)| + |\lambda(w)|$ is at most $4d\,\|\lambda\|^2$,

$$|H_U(B, D_U\lambda)| \le c_{\mathrm{br}}\sqrt{d}\, j_*\, \|B\|\, \|\lambda\|\,.$$

At unit spacing, the change of the quadratic form obeys

$$\begin{aligned}|H_U(B + D_U\lambda, B + D_U\lambda) - H_U(B,B)| \le{}& 2c_{\mathrm{br}}\sqrt{d}\, j_*\, \|B\|\, \|\lambda\| \\ &+ 2c_{\mathrm{br}} d\, j_*\, \|\lambda\|^2\,.\end{aligned}$$

Thus no derivative of the current is required for this correction. Complex backgrounds require explicit bounds on their adjoint transports. An arbitrary independent current is not automatically the derivative of an invariant action; its analytic extension must be checked through the actual operator formula. Raw II, lines 9380–9560, already gives the corresponding physical Ward calculation with the opposite gauge convention.

There is an even sharper physical-fiber statement. If the averaging level set is preserved by rooted gauges and the field is a critical point of the action restricted to that level set, the complete Hessian in a level-set chart annihilates rooted gauge directions. The mixed derivative of gauge invariance equals the Hessian pairing plus a first derivative applied to the mixed chart term; the latter is zero by constrained criticality. This includes the constraint-curvature/current correction. It supplies no analogous identity for a freely prescribed current and no axial gap by itself.

4 Interaction smallness at the larger real cutoff

Lemma 4.1 (Physical-field Schwarz estimate)**.** *Let $F_Y(A,z)$ be holomorphic on the balanced physical-field ball $\|z\|_{\infty,Y} < r_0$, uniformly over its stated background domain. Assume $|F_Y| \le a_Y$ and $D_z^j F_Y(A,0) = 0$ for $j = 0,1,2$. Then*

$$|F_Y(A,z)| \le a_Y r_0^{-3} \|z\|_{\infty,Y}^3 .$$

For real B, $\|B\|_{\infty,Y} \le R$, and $gR < r_0$, the action remainder $W_Y = g^{-2} F_Y(A, gB)$ satisfies

$$|W_Y| \le \frac{a_Y}{r_0^3} gR \|B\|_{2,Y}^2 , \qquad \sup_{\text{cutoff}} |W_Y| \le \frac{a_Y}{r_0^3} gR^3. \tag{11}$$

Proof. For $z \neq 0$, apply Schwarz's lemma three times to $w \mapsto F_Y(A, r_0 wz/\|z\|_\infty)/a_Y$ on the unit disk. The zero cases and non-strict limits are immediate. Substitute $z = gB$ and use $\|B\|_\infty^3 \le R\|B\|_2^2$. □

If $\sup_b \sum_{Y \ni b} a_Y \le a_*$, summing gives

$$\sum_Y |W_Y| \le a_* gR r_0^{-3} \sum_b |B_b|^2.$$

This can be absorbed into a fixed real Gaussian quadratic exponent when gR is sufficiently small. More generally, a term $g^{-r} F_Y(gB)$ vanishing to order m has cutoff norm at most $a_Y g^{m-r} R^m / r_0^m$. A local norm $\sup_b \sum_{Y \ni b} e^{a|Y| + \kappa \ell(Y)} a_Y$ transfers with the same factor. An extensive bound on the sum of the functions does not imply such a local norm. A physical argument such as gCB or a nonlinear minimizer response needs its own uniform bound and local support expansion before this lemma can be applied.

4.1 The full exponent is not cubic

The full fluctuation potential in CMP109, pages 266–270, and Raw II, lines 73910–73940, also includes change-of-variables and Haar factors. Its schematic orders in the physical argument $z = gCB$ are:

Term	First order	Cutoff scale
Logarithm of the Jacobian	1	gR
Haar log density	2	g^2R^2
Action, gauge, and current cubic remainders	3 with g^{-2}	gR^3
Quadratic term in the quadratic constraint correction	4 with g^{-2}	g^2R^4

These orders use the stated cancellations and do not establish their localized coefficient bounds. Previous effective-action increments are separate induction inputs. For $R \ge 1$ and $gR \le 1$, all displayed scales are bounded by a constant times gR^3. Thus such coefficient bounds imply $u_Y := \sup_{\text{cutoff}} |V_Y| \le C_Y gR^3$. For $u_Y \le 1$,

$$v_Y := \sup_{\text{cutoff}} |e^{V_Y} - 1| \le e^{u_Y} - 1 \le 2u_Y. \tag{12}$$

A nonzero linear term cannot obey $|V_Y(B)| \le CgR\|B\|^2$ near zero. An additive constant is necessary, for example $Cg\|B\| \le (Cg/2)(1+\|B\|^2)$. The above orders allow a safe envelope $\widetilde{C}_Y g(1+R)(1+\|B\|_{2,Y}^2)$, with summable localized constants. Odd Gaussian cancellations do not repair a false pointwise quadratic bound.

5 A complete interaction theorem for disjoint cells

Let Λ be a finite set of cells of fixed real dimension D, with metric d. Suppose their holomorphic complex symmetric covariance obeys

$$\operatorname{Herm} C(A) \geq \lambda I, \qquad \|C(A)\| \leq \Lambda_0, \qquad \|C_{ij}(A)\| \leq J_{ij} = J_{ji},$$

$$A_\kappa := \sup_i \sum_{j \neq i} e^{\kappa d(i,j)} J_{ij} < \infty.$$

Each interaction $V_i(A, x_i)$ reads only its own cell's fluctuation. It is measurable in real x_i, holomorphic in the background on the fixed domain, and obeys

$$\sup_{A, |x_i| \leq R} |e^{V_i(A,x_i)} - 1| \leq v.$$

Put $\chi_i = \mathbf{1}_{\{|x_i| \leq R\}}$ and define globally

$$q_i = -\mathbf{1}_{\{|x_i| > R\}} + \chi_i (e^{V_i} - 1).$$

The values of V_i outside the cutoff are unnecessary. Then

$$Z_V(A) = \mathbb{E}_{C(A)} \prod_i (1 + q_i) = \mathbb{E}_{C(A)} \left(\prod_i \chi_i \right) e^{\sum_i V_i}.$$

The complex Gaussian uses the normalized analytic determinant branch on the accretive cone. Let

$$\delta = \lambda/2, \quad V_* = 8\Lambda_0^2/\lambda, \quad K_G = (2V_*/\lambda)^{D/2}, \quad p_R = 2^{D/2} e^{-R^2/(4V_*)},$$
$$\beta = K_G(p_R + v), \quad B_0 = 8e^2 A_\kappa/\delta, \quad \beta_* = [2e^2 \max(1, B_0)]^{-1}.$$

Here $\ell(X)$ denotes minimal spanning-tree length in the cell metric.

Theorem 5.1. *If $\beta \leq \beta_*$, there is a convergent analytic logarithm*

$$\log Z_V(A) = \sum_{\varnothing \neq X \subset \Lambda} U_V(X; A),$$

whose canonical forest/Mayer activities satisfy

$$\sup_i \sum_{X \ni i} e^{\kappa \ell(X) + |X|} \|U_V(X; \cdot)\|_\infty \leq 2e^2 \beta. \tag{13}$$

The constants are independent of volume. The set X is a set of marked fluctuation cells; it is not yet a claim of background support.

Proof. BKAR partition-mask covariances are convex combinations of block principal submatrices, and preserve the stated accretivity and norm bounds. Allow a further operator perturbation of norm at most δ. The resulting M has $\operatorname{Herm} M \geq \lambda I/2$, $\|M\| \leq 2\Lambda_0$, $\operatorname{Herm} M^{-1} \geq V_*^{-1} I$, and $|\det M| \geq (\lambda/2)^{D|S|}$. Its density modulus is bounded by $K_G^{|S|}$ times a product of real cell Gaussian densities of covariance $V_* I$. Hence

$$\left| \int p_M(x_S) \prod_{i \in S} q_i(A, x_i)\, dx_S \right| \leq K_G^{|S|} \prod_{i \in S} \mathbb{E}_{V_* I}(\mathbf{1}_{\{|G_i| > R\}} + v) \leq \beta^{|S|}.$$

The tail estimate is exponential Markov with parameter $1/(4V_*)$. The same domination establishes background holomorphy.

For a tree T with degrees d_i, independent covariance-edge parameter changes of radii $r_{ij} = \delta/[\max(d_i, d_j)J_{ij}]$ have every block row and column perturbation bounded by δ. Cauchy's formula and $d^d \le e^d d!$ give

$$|\partial_T f_S| \le \beta^{|S|}(e^2/\delta)^{|S|-1} \prod_i d_i! \prod_{ij\in T} J_{ij}.$$

The multipliers are independent of the covariance interpolation parameters. At a disconnected forest mask their product expectation factorizes over the components. BKAR thus yields a hard-core gas of disjoint marked sets with the exact tree activities.

For $n \ge 2$ the degree sum is

$$\sum_{T \text{ on } [n]} \prod_i d_i! = (n-2)!\binom{3n-3}{n-2}, \qquad \frac{1}{(n-1)!}\sum_T \prod_i d_i! \le 8^{n-1}.$$

Summing leaves with the weighted J norm proves

$$\sup_i \sum_{S\ni i} \|\zeta(S)\|_\infty \, e^{\kappa\ell(S)+\alpha|S|} \le \frac{\beta e^\alpha}{1 - B_0\beta e^\alpha}.$$

At $\alpha = 2$, the stated ceiling implies the Kotecký–Preiss criterion for activities weighted by $e^{\kappa\ell(S)+|S|}$ and size function $|S|$. The rooted Mayer bound, grouped by the union of its overlapping marked polymers, proves (13). This is precisely the proof of Notebook 03 with the marked-tail parameter replaced by $K_G(p_R + v)$. □

Corollary 5.2 (The interaction changes the logarithm by $O(v)$)**.** *If $\tau := K_G p_R \le \beta_*/4$ and $K_G v \le \beta_*/4$, the same canonical activities for V and for $V = 0$ obey*

$$\sup_i \sum_{X\ni i} e^{\kappa\ell(X)+|X|} \, \|U_V(X) - U_0(X)\|_\infty \le 8e^2 K_G v. \tag{14}$$

Proof. Interpolate the bounded multipliers by $q_i(t) = -\mathbf{1}_{\mathrm{bad},i} + t\chi_i(e^{V_i} - 1)$. For $v > 0$, the disk $|t| \le \rho = \beta_*/(2K_G v) \ge 2$ has parameter at most $3\beta_*/4$. The activity vector is analytic there in the weighted norm and bounded by $M = 2e^2\beta_*$. Banach-valued Cauchy bounds for its Taylor coefficients give

$$\|U(1) - U(0)\| \le \frac{M}{\rho - 1} \le 8e^2 K_G v.$$

The $v = 0$ case is immediate. Both logarithms use the expansion's branches; no lower bound for a cutoff probability is used. □

5.1 The strict-background-local version

Retain *all* hypotheses and smallness conditions of Notebook 04, including its fixed background-independent baseline, local precision decomposition, product domain, and constants. Require additionally that V_i reads only background coordinates in cell i. In the rooted support algebra for a marked set S, its extra multiplier has support $\{i\} \subset S$, zero added support length, and norm at most v. Consequently the multiplier product norm is bounded pointwise by $\prod_{i\in S}(\mathbf{1}_{\mathrm{bad},i} + v)$.

The marginal-density estimate of that notebook is

$$\|p_M(x_S)\|_{\mathcal{B}_S} \le \widehat{K}_G^{|S|} p_{2V_* I}(x_S), \qquad \widehat{K}_G = (8V_*/\lambda)^{D/2}.$$

Its proof therefore goes through with the replacement

$$\widehat{\beta} = \widehat{K}_G[2^{D/2}e^{-R^2/(8V_*)} + v].$$

Use that notebook's $\delta = \lambda/4$, edge envelope, B_0, and ceiling $\widehat{\beta} \le [2e^2 \max(1, B_0)]^{-1}$. Forest factorization is still valid coefficientwise in the local source variables, and compatibility is disjointness of the marks, not of their added background supports. Its connected-hull construction then gives activities $E_{\Lambda,H}(A|_H)$ with

$$\sup_b \sum_{H \ni b} e^{a_*|H|} \left\| E_{\Lambda,H} \right\|_\infty \le 2e^2 S_{\kappa/2} \widehat{\beta}, \qquad a_* = \min(1/2, \kappa/4),$$

where $S_{\kappa/2} = \sum_{z \in \mathbb{Z}^d} e^{-(\kappa/2)|z|}$. Under the analogous quarter budgets, the difference from the Gaussian-cutoff activities is at most $8e^2 S_{\kappa/2} \widehat{K_G} v$ in this norm.

This statement would be false without controlling the interaction's background support. An interaction in cell i that reads an arbitrarily remote background variable cannot be assigned support $\{i\}$. Nor does strict dependence on the varying background imply consistency between volumes: the finite-volume coefficients may still depend on the fixed whole-volume baseline. The symmetry and domain caveats of Notebook 04 remain in force.

6 The actual enlarged domain and the remaining interaction theorem

Write $L_g = \log g^{-2}$, $R = A_1 L_g^{p_0}$ and $P = C_1 L_g^{p_1}$ with $p_1 < p_0$. For every fixed q,

$$gR^q = A_1^q e^{-L_g/2} L_g^{qp_0} \longrightarrow 0.$$

Thus a fixed physical-field analytic radius permits the larger real cutoff, after separately enforcing $gR < r_0$. It does not require $R \le P$, which in fact fails for small g. A hypothesis $gP < r_0$ alone is insufficient; a domain only of unscaled radius P cannot be enlarged by Schwarz's lemma. After the primitive constants are fixed, choose the coupling ceiling last to meet the cluster budget and any prescribed absolute error. The constant in $O(gR^3)$ is independent of that later target; the ceiling need not be.

When $p_0 > 1/2$, the Gaussian tail Ke^{-cR^2} is smaller than every fixed gL_g^q scale for sufficiently small g, since $-cA_1^2 L_g^{2p_0} + L_g/2 - q \log L_g \to -\infty$.

The deposit already contains the appropriate larger-radius calculation. Raw II, `5.2:lem-complex-localisation`, proof Step 3, lines 11178–11195, bounds cubic pieces by $C(M, \kappa_1) g_k (T^{\mathbb{C}})^3 e^{-\kappa_V d_k(Y)}$, treats merely vanishing pieces by a $g_k T^{\mathbb{C}}$ factor, and chooses the coupling ceiling after the activity target. Its domain calculation uses

$$\theta_k^{\mathbb{C}} = K_R g_k T^{\mathbb{C}} / \alpha_{*,k}, \qquad \mathsf{w}^{\mathbb{C}} = 1/\theta_k^{\mathbb{C}} - 1, \qquad (1 + \mathsf{w}^{\mathbb{C}}) K_R g_k T^{\mathbb{C}} = \alpha_{*,k}.$$

It invokes physical response containment; it does not claim that the larger logarithmic radius fits inside the smaller one. The remaining verification is its imported response theorem, complex background chart, admitted histories, and summable localized coefficient bounds. Raw II's later box-exponent lemma also assumes a coupling-flow hypothesis, so it cannot independently prove that same hypothesis.

6.1 Why overlapping interactions need more than the cell theorem

For a potential $\sum_Y V_Y(A, B_Y)$ with overlapping cell sets, define $\chi_Y = \prod_{i \in Y} \chi_i$ and $w_Y = \chi_Y(e^{V_Y} - 1)$, setting $w_Y = 0$ outside its cutoff. There is an exact identity

$$\left(\prod_i \chi_i \right) e^{\sum_Y V_Y} = \left(\prod_i \chi_i \right) \prod_Y (1 + w_Y), \qquad \|w_Y\|_\infty \le e^{u_Y} - 1.$$

If any cutoff fails, both sides vanish; otherwise the identity is the product of exponentials. A bad mark in Y also annihilates a product containing w_Y.

These facts do not imply Theorem 5.1 for overlapping interactions. A covariance mask fails to factorize across a partition cut by an interaction hyperedge. Grouping overlapping interactions into a molecule S makes a Gaussian block of dimension $D|S|$. A crude edge bound can then pay $|S|$ at every incident edge, hence $|S|^{d_S}$ at degree d_S. The degree factorials from Cauchy's formula can consume the factorial savings in tree enumeration. One exponential factor in $|S|$ does not control unrestricted powers of $|S|$ in that argument. Merely calling these molecules "cells" therefore does not prove convergence.

A natural prospective hypothesis is a small weighted norm

$$\sup_i \sum_{Y \ni i} e^{a|Y| + \kappa \ell(Y)} (e^{u_Y} - 1),$$

with enough reserve for dimensional factors. No theorem that this hypothesis alone completes the proposed molecule argument is claimed here. One needs a valid allocation of forest edges to fixed cells, or an interaction-specific analytic expansion. CMP116, pages 10–11 and 16–17, uses a field-dependent quadratic-form decomposition and a separate Gaussian/determinant resummation. The next precise model target is to verify that theorem on the enlarged physical domain, with its actual complex operators and history norm.

7 What these results contribute to the construction

The flat gauge bridge uses the genuine gauge penalty and both genuine constraints. It can transfer a matched axial estimate to the Gaussian needed for the cutoff calculation, with explicit coordinate constants. Companion Notebook 06 now supplies a direct flat axial gap and the complementary energy upper bound, using curvature contraction and a compact right inverse to averaging. This replaces the imported flat spectral input for the whole-lattice or periodic quadratic model; the actual smooth minimizer/constraint identification and complex operator estimates remain explicit. The physical-field estimates explain how the larger cutoff can be handled and identify the linear term that a cubic-only argument misses. The fixed-cell theorem closes a complete, limited interaction module.

For the scalar quadratic cutoff coefficient, Raw I, lines 3780–3837, extracts the carrier Hessian at $(U, J) = (1, 0)$ followed by normalized responses and a moment sum. A genuinely flat baseline can therefore be sufficient for the direct-Hessian route of Notebook 03. Choosing a different baseline $C_0(U_{\mathrm{ref}})$ at every real reference only localizes perturbation dependence; it leaves arbitrary dependence on U_{ref} and proves no full-background locality.

Still required are uniform covariance/response bounds for the actual independent-current family, the overlapping interaction estimate, and their match to the effective-action induction. Beyond this local RG step, continuum compactness, a sufficiently rich observable hierarchy, rotation restoration, Yang–Mills identification, nontriviality, and a mass gap remain separate obligations listed in Notebook 01.

References

[1] T. Bałaban, *Propagators for lattice gauge theories in a background field*, Commun. Math. Phys. 99 (1985), 389–434. Publisher record.

[2] T. Bałaban, *Renormalization group approach to lattice gauge field theories I*, Commun. Math. Phys. 109 (1987), 249–301. Copy of the original article.

[3] T. Bałaban, *Renormalization group approach to lattice gauge field theories II. Cluster expansions*, Commun. Math. Phys. 116 (1988), 1–22. Copy of the original article.

[4] A. Abdesselam, Author-hosted notes on the BKAR formula.

[5] R. Kotecký and D. Preiss, *Cluster expansion for abstract polymer models*, Commun. Math. Phys. 103 (1986), 491–498. Publisher record.